Formula of X-grade chapters

(a) Real numbers

1) Euclid's Division Lemma

$$a = bq + r \qquad 0 \leq r < b$$

2) L.C.M × H.C.F = a × b

3) $\frac{p}{q}$

if $q = 2^n 5^m$ then $\frac{p}{q}$ is terminating decimal

if $q \neq 2^n 5^m$ then $\frac{p}{q}$ is non-terminating repeating decimal.

(b) Polynomials

1) $ax^2 + bx + c$ → α, β are zeros

$\alpha + \beta$ = sum of zeros = $-\frac{b}{a}$

$\alpha\beta$ = Product of zeros = $\frac{c}{a}$

2) If α and β are zeros of polynomial, then quadratic poly. is $x^2 - (\alpha+\beta)x + \alpha\beta$

i.e $x^2 - (\text{Sum})x + \text{Product}$

③ $ax^3 + bx^2 + cx + d$ → α, β, γ

$\alpha + \beta + \gamma = \frac{-b}{a}$

$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$

$\alpha\beta\gamma = -\frac{d}{a}$

④ If α, β, γ are zeros of cubic poly. then poly. is $x^3 - (\alpha+\beta+\gamma)x^2 + (\alpha\beta + \beta\gamma + \gamma\alpha)x - \alpha\beta\gamma$.

⑤ Dividend = Divisor X Quitient + Remainder

If remainder = 0
then Divisor is a factor of Dividend.

$P(x) = g(x) \times q(x) + r(x)$

⑥ Some Algebraic helpful for Polynomial chapter

(i) $(\alpha + \beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta$

(ii) $(\alpha - \beta)^2 = \alpha^2 + \beta^2 - 2\alpha\beta$

(iii) $(\alpha^2 - \beta^2) = (\alpha - \beta)(\alpha + \beta)$

(iv) $\alpha^3 + \beta^3 = (\alpha + \beta)(\alpha^2 - \alpha\beta + \beta^2)$

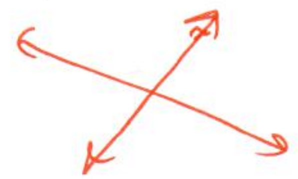
(v) $\alpha^3 - \beta^3 = (\alpha - \beta)(\alpha^2 + \alpha\beta + \beta^2)$

(c) linear equation in two variables

(1) $a_1x + b_1y + c_1 = 0$

$a_2x + b_2y + c_2 = 0$

* $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$

* 

* unique solution
* Consistent solution
* intersecting lines

$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

* infinite many solution
* consistent solution
* overlapping lines

$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

* No solution
* inconsistent solution
* Parallel lines

(2) learn three methods, → elimination, substitution, cross-multiplica

(d) Quadratic Equation.

(1) learn factorisation method (middle-term splitting), completing the square, Quadratic formula.

(2) $ax^2 + bx + c = 0$

Discriminant, $D = b^2 - 4ac$

$$x = \frac{-b \pm \sqrt{D}}{2a}$$

i.e $x = \frac{-b + \sqrt{D}}{2a}, \frac{-b - \sqrt{D}}{2a}$

③ Nature of roots:-

$$D = b^2 - 4ac$$

if $D > 0$ Roots are real and unequal

$D = 0$ Roots are real and equal

$D < 0$ Roots are unreal.

ⓔ Arithmetic progression

① A.P :-

$$a, \ a+d, \ a+2d + \ldots\ldots$$

② $a_n = a + (n-1)d$

③ $S_n = \frac{n}{2}\left[2a + (n-1)d\right]$

④ $S_n = \frac{n}{2}\left[a + a_n\right]$ → used when 'a' and 'an' both are given.

⑤ If a, b, c be consecutive terms of A.P

then $b - a = c - b$

$b + b = a + c$

$$\boxed{2b = a + c}$$

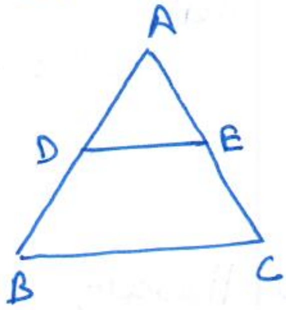
⑥ Any three consecutive terms in A.P is taken as

$$a-d, \ a, \ a+d.$$

⑦ $a_n = S_n - S_{n-1}$.

(f) Triangles

(1) BPT theorem



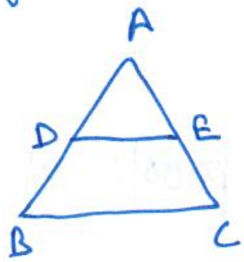
if $DE \parallel BC$

then $\dfrac{AD}{BD} = \dfrac{AE}{EC}$

or

$\dfrac{AD}{AB} = \dfrac{AE}{AC}$

(2) Converse of BPT



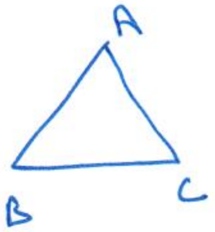
If $\dfrac{AD}{BD} = \dfrac{AE}{EC}$

or

$\dfrac{AD}{AB} = \dfrac{AE}{AC}$

then $DE \parallel BC$

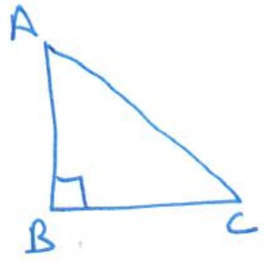
(3) The ratio of areas of two similar Δs is equal to the square of ratio of corresponding sides.



if $\Delta ABC \sim \Delta PQR$

then $\dfrac{ar(ABC)}{ar(PQR)} = \left(\dfrac{AB}{PQ}\right)^2 = \left(\dfrac{BC}{QR}\right)^2 = \left(\dfrac{AC}{PR}\right)^2$

(4) Pythagoras theorem,



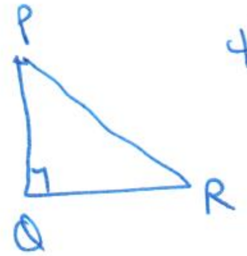
if $\angle B = 90^\circ$

then

$$AC^2 = AB^2 + BC^2$$

(5) Converse of pythagoras theorem,

if $PR^2 = QR^2 + PQ^2$



then $\angle Q = 90^\circ$

(6) Learn statements and proofs of all 4 theorems.

(9) Co-ordinate Geometry

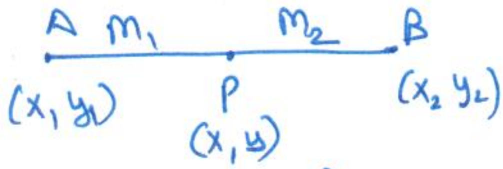
(1) Distance formula

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

(2) Section formula

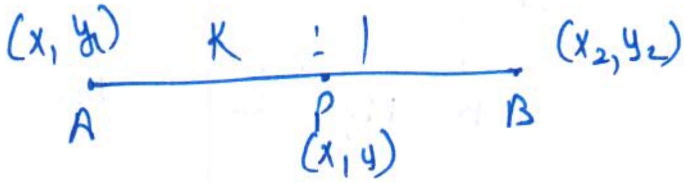
If ratio is known

$$\frac{AP}{BP} = \frac{m_1}{m_2}$$



$$P \equiv \left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$$

If you want to find ratio
then use $K:1$

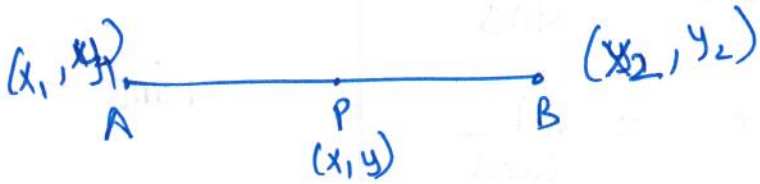


$$P \equiv \left(\frac{Kx_2 + x_1}{K+1} , \frac{Ky_2 + 1}{K+1} \right)$$

(3) A point on x-axis is $(x, 0)$

(4) A point on y-axis is $(0, y)$

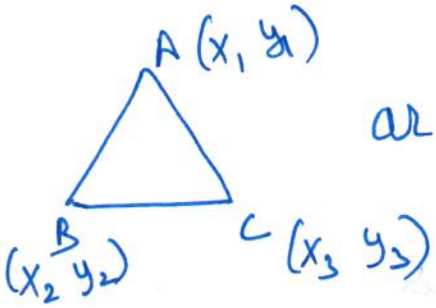
(5) mid-point formula



if $AP = BP$

then $$P \equiv \left(\frac{x_1 + x_2}{2} , \frac{y_1 + y_2}{2} \right)$$

(6)

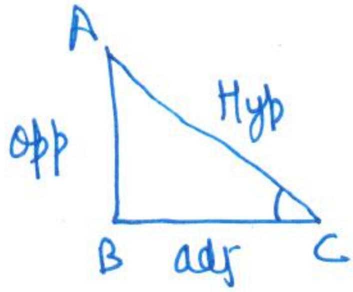


$$ar(ABC) = \left| \frac{1}{2}\left[x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2) \right] \right.$$

(7) If A, B, C points are collinear,
then $ar(ABC) = 0$.

(8) Introduction to trigonometry

(1)



$$\sin C = \frac{\text{opp}}{\text{Hyp}}$$

$$\cos C = \frac{\text{adj}}{\text{Hyp}}$$

$$\tan C = \frac{\text{opp}}{\text{adj}}$$

$$\operatorname{cosec} C = \frac{\text{Hyp}}{\text{opp}}$$

$$\sec C = \frac{\text{Hyp}}{\text{adj}}$$

$$\cot C = \frac{\text{adj}}{\text{opp}}$$

(2)

$$\operatorname{cosec}\theta = \frac{1}{\sin\theta}$$

$$\sec\theta = \frac{1}{\cos\theta}$$

$$\tan\theta = \frac{1}{\cot\theta}$$

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$

$$\cot\theta = \frac{\cos\theta}{\sin\theta}$$

(3)

$$\sin(90-\theta) = \cos\theta$$

$$\cos(90-\theta) = \sin\theta$$

$$\sec(90-\theta) = \operatorname{cosec}\theta$$

$$\operatorname{cosec}(90-\theta) = \sec\theta$$

$$\cot(90-\theta) = \tan\theta$$

$$\tan(90-\theta) = \cot\theta$$

(4)

$$\left.\begin{aligned} \sin^2 A + \cos^2 A &= 1 \\ \sin^2 A &= 1 - \cos^2 A \\ \cos^2 A &= 1 - \sin^2 A \end{aligned}\right\}$$

$$\left.\begin{aligned} \sec^2 A &= 1 + \tan^2 A \\ \sec^2 A - 1 &= \tan^2 A \\ \sec^2 A - \tan^2 A &= 1 \end{aligned}\right\}$$

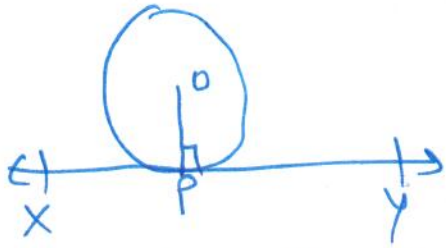
$$\left.\begin{aligned} \operatorname{cosec}^2 A &= 1 + \cot^2 A \\ \operatorname{cosec}^2 A - 1 &= \cot^2 A \\ \operatorname{cosec}^2 A - \cot^2 A &= 1 \end{aligned}\right\}$$

(5)

	$\sqrt{\frac{0}{4}}$	$\sqrt{\frac{1}{4}}$	$\sqrt{\frac{2}{4}}$	$\sqrt{\frac{3}{4}}$	$\sqrt{\frac{4}{4}}$
	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
Tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	N.D
cosec	N.D	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
sec	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	N.D
cot	N.D	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

(h) Circles

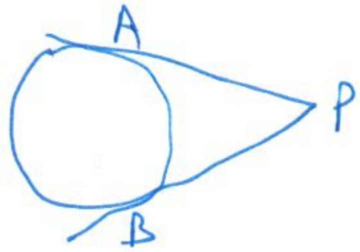
(1) Radius is perp. to tangent



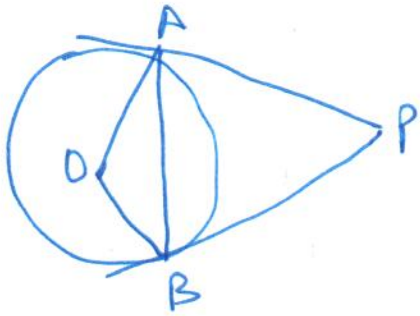
$OP \perp XY$

(2) The length of tangents drawn from external pt to circle are equal.

$PA = PB$

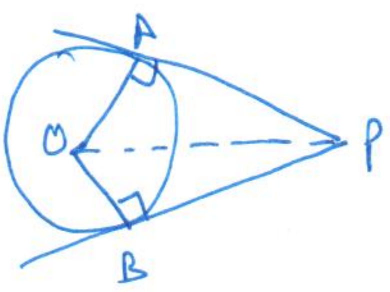


(3)



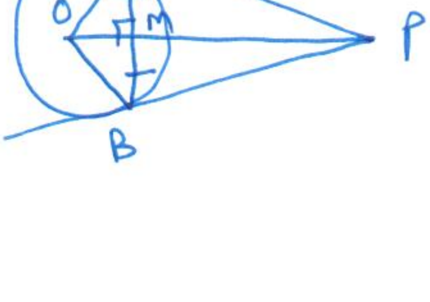
$2\angle OAB = \angle APB$

(4)



$\angle APO = \angle BPO$

(5)



$OP \perp AB$ and $AM = BM$

(i) Areas related to circles

(1)

Circumference = Perimeter of circle = $2\pi r$

Area $= \pi r^2$

(2)

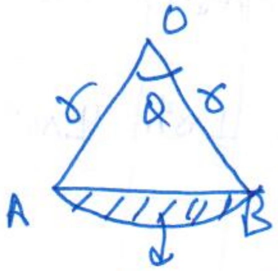
Area of minor sector

$= \frac{\theta}{360} \pi r^2$

and

area of major sector $= \pi r^2 - \frac{\theta}{360} \pi r^2$

(3)



minor segment

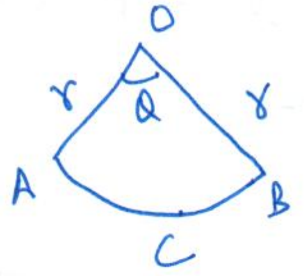
area of minor segment

$= \frac{\theta}{360} \pi r^2 -$ area of ΔAOB

and

area of major segment $= \pi r^2 -$ area of minor segment

(4)



Perimeter of minor sector

$= OA + \text{arc}(ACB) + OB$

$= r + \frac{\theta}{360}(2\pi r) + r$

Perimeter of sector $= 2r + \frac{\theta}{360} \cdot 2\pi r$

(5)

minor arc

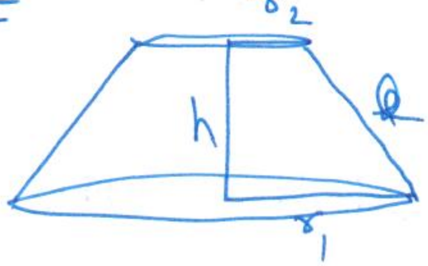
lenth of minor arc $= \frac{\theta}{360} \cdot 2\pi r$

(j) Surface areas and volumes

	Cuboid	Cube	Cylinder	Cone	Sphere	hemisphere
C.S.A	$2(l+b)h$	$4a^2$	$2\pi rh$	πrl		$2\pi r^2$
T.S.A	$2(lb+bh+hl)$	$6a^2$	$2\pi r(h+r)$	$\pi r(l+r)$	$4\pi r^2$	$3\pi r^2$
volume	$l \times b \times h$	a^3	$\pi r^2 h$	$\frac{1}{3}\pi r^2 h$	$\frac{4}{3}\pi r^3$	$\frac{2}{3}\pi r^3$

$l^2 = r^2 + h^2$

frustum of cone



$l^2 = h^2 + (r_1 - r_2)^2$

C.S.A $= \pi(r_1 + r_2)l$

T.S.A $= \pi(r_1 + r_2)l + \pi r_1^2 + \pi r_2^2$

volume $= \frac{1}{3}\pi(r_1^2 + r_2^2 + r_1 r_2)h$

$$1\,m^3 = 1000\,l$$

$$1\,l = 1000\,cm^3$$

$$1\,m = 100\,cm$$

$$1\,dm = 10\,cm$$

$$1\,m = 10\,dm$$

(K) **Statistics**

① **Mean**

Direct method :- $\overline{x} = \dfrac{\Sigma f_i x_i}{\Sigma f_i}$

assumed mean method :- $\overline{x} = a + \dfrac{\Sigma f_i d_i}{\Sigma f_i}$

where $d_i = x_i - a$

step-deviation method :-

$$\overline{x} = a + \frac{\Sigma f_i u_i}{\Sigma f_i} \times h$$

where $u_i = \dfrac{x_i - a}{h}$

②

$$\text{mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

③ median

$$\text{median} = l + \left(\frac{\frac{n}{2} - C.F}{F}\right) \times h$$

④ Empirical formula (3,2)

$$3\,\text{median} = \text{mode} + 2\,\text{mean}$$

(l) Probability

① $P(E) = \dfrac{\text{favourable outcomes to } E}{\text{total no of outcomes}}$

② $P(\text{not } E) = 1 - P(E)$

③ $0 \le P(E) \le 1$